- *Quilc:* A fast compiler that takes a Quil (Quantum Instruction Language) program as input and outputs gate instructions for Rigetti's hardware (in an optimised way).

Forest SDK is designed to facilitate seamless integration with Rigetti Quantum Cloud Services (QCS), which allow developers to evaluate quantum algorithms on real superconducting qubit devices and simulators. The SDK's modular structure fits both academic research and industry-based prototyping.

Use cases

- *Quantum implementation development:* Using pyQuil to build and compile circuits while exploring algorithms for implementation.
- *Real hardware execution:* Running workloads on Rigetti's quantum processors through QCS.
- *Hybrid quantum-classical workflows:* Using quantum circuits in conjunction with classical resources for variational and optimisation algorithms.
- *Compiling research:* Using Quilc to evaluate and enhance quantum compilation.
- *Education and prototyping:* Allowing quantum algorithm development in the early stages in educational and research environments.
- *Applied quantum algorithms:* Support for simulations and optimisation problems in finance, manufacturing, and logistics.

With its simulators, Forest SDK affords a practical testbed for innovators looking to connect today's world of quantum research with tomorrow's commercial reality.

Ocean SDK

D-Wave's Ocean is a quantum software development kit specifically designed for quantum annealing, which is quite different from gate-based computing. Rather than translate universal quantum logic into quantum code, Ocean is aimed at solving combinatorial optimisation problems by mapping into Quadratic Unconstrained Binary Optimisation (QUBO) models.

The SDK leverages Python along with a variety of tools, and is of great importance to businesses that grapple with optimisation challenges on a large scale. It helps integrate with D-Wave's Leap cloud platform and provides full access to quantum annealers.

Use cases

- *Logistics optimisation:* Improving route planning, vehicle scheduling, and warehouse management.
- *Supply chain management:* Overcoming bottlenecks throughout production, inventory and distribution networks.
- *Portfolio optimisation:* Assisting financial institutions to manage investment risks and returns.
- *Scheduling:* Controlling workforce shifts, manufacturing order, and project planning.
- *Energy sector optimisation:* Ramping up grid balancing, renewable energy planning, and resource allocation.
- *Combinatorial optimisation research:* Quantifying quantum annealing's performance against classical optimisation.
- *AI and ML acceleration:* Using annealing to train Boltzmann machines and generative models.

ProjectQ

ProjectQ is a minimal, open source quantum framework developed with the aim of being simple and modular. Written in Python, it provides researchers and educators the ability to rapidly prototype quantum algorithms and experiment with ideas without worrying about heavy dependencies.

The ProjectQ framework allows researchers to implement simulation backends, execute experiments on IBM Quantum hardware, and structure projects to easily switch from theory to practice. ProjectQ's modular design lends itself for easy extensions – it's an excellent framework for research, teaching, and proof-of-concept demonstrations.

Use cases

- *Prototyping quantum algorithms:* Quickly testing and iterating on an algorithmic idea.
- *Education/training:* Familiarising both faculty and students with otherwise complicated quantum ideas.
- *Integration with IBM hardware:* Executing circuits on real hardware.
- *Modular design for research:* Allowing researchers to extend the platform when they want to run

Table 1: A comparison of popular open source frameworks for quantum computing

Framework	Hardware support	Specialisation	Target audience
Qiskit	IBM quantum devices	General-purpose (gate model)	Broad developer base
Cirq	Google Sycamore	NISQ circuits, error studies	Researchers, academics
PennyLane	Multiple backends	Hybrid quantum-classical ML	AI/ML researchers
Forest SDK	Rigetti QCS	Gate model with Quil	Developers, educators
Ocean SDK	D-Wave annealers	Optimisation (QUBO)	Industry practitioners
ProjectQ	Simulators + IBM	Educational, prototyping	Academics, students